SECTION B

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
8				**Indicative content** use insulated container add known volume of HCl / known concentration excess to ensure that all CaO reacts measure initial temperature of HCl add known mass of CaO and stir well for rapid and complete reaction measure temperature change at intervals plot temperature change over time and extrapolate lines determine temperature increase repeat all steps using $Ca(OH)_2$ calculate number of moles of CaO / $Ca(OH)_2$ reacted $n = \frac{mass}{M_r}$ calculate enthalpy change ΔH_2 and ΔH_3 $\Delta H = -\frac{mc\Delta T}{n}$ use Hess's cycle to calculate ΔH_1 $\Delta H_1 = \Delta H_2 - \Delta H_3$	4		2	6	2	4

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
				5-6 marks Good description of method and steps in calculation *There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured. The information included in the response is relevant to the argument.* **3-4 marks** Basic reference to most steps in method and part of calculation *There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure. Mainly relevant information is included in the response but there may be some minor errors or the inclusion of some information not relevant to the argument.* **1-2 marks** Reference to some steps in method or part of calculation *There is a basic line of reasoning which is not coherent, supported by limited evidence and with very little structure. There may be significant errors or the inclusion of information not relevant to the argument.* **0 marks** No attempt made or no response worthy of credit.						
				Question 8 total	**4**	**0**	**2**	**6**	**2**	**4**